WORKED SOLUTION

STEP II 2005, Question 10

Mechanics • Colliding Projectiles

STEP II 2005, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q10

The key idea

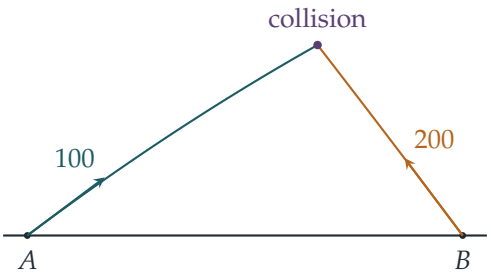
Both missiles are projectiles, so we can write down where each one is at any time. A collision means the two positions agree at the *same* instant, which gives two equations (horizontal and vertical) in two unknowns: the collision time and T. The horizontal equation is linear, so it lets us eliminate the collision time, and the vertical one then becomes a quadratic in T. We then have to decide which root makes physical sense.

Solution

Setting up

We model both missiles as particles moving freely under gravity, with no air resistance. Take A as the origin, the x-axis along AB (so B is at $x = 180$) and the y-axis vertically upwards. Measure time t in seconds from the launch of the first missile.

For the first missile, $\sin\alpha = \frac{3}{5}$ gives $\cos\alpha = \frac{4}{5}$, so its initial velocity components are $100 \cdot \frac{4}{5} = 80$ and $100 \cdot \frac{3}{5} = 60$. For the second, $\sin\beta = \frac{4}{5}$ gives $\cos\beta = \frac{3}{5}$. It travels towards A, so its components are $-200 \cdot \frac{3}{5} = -120$ and $200 \cdot \frac{4}{5} = 160$.



Let $s = t - T$ be the time since the second launch. Then, while both are in flight,

$$\begin{aligned} &\text{first missile:} && x = 80t, && y = 60t - 5t^2, \\ &\text{second missile:} && x = 180 - 120s, && y = 160s - 5s^2. \end{aligned}$$

The collision

At the collision the x-coordinates agree:

$$80t = 180 - 120(t - T) \quad \Longrightarrow \quad 200t = 180 + 120T \quad \Longrightarrow \quad t = 0.9 + 0.6T,$$

and so $s = t - T = 0.9 - 0.4T$.

The y-coordinates also agree: $60t - 5t^2 = 160s - 5s^2$, that is,

$$60t - 160s = 5(t^2 - s^2) = 5(t - s)(t + s) = 5T(t + s),$$

using $t - s = T$. Substituting the expressions for t and s:

$$\begin{aligned} 60t - 160s &= 60(0.9 + 0.6T) - 160(0.9 - 0.4T) = 100T - 90, \\ 5T(t + s) &= 5T(1.8 + 0.2T) = 9T + T^2. \end{aligned}$$

Equating these,

$$T^2 - 91T + 90 = 0 \quad \Longrightarrow \quad (T - 1)(T - 90) = 0,$$

so $T = 1$ or $T = 90$.

Ask yourself Both roots solve the equations. Why can't $T = 90$ be the answer?

For $T = 90$ we get $s = 0.9 - 36 = -35.1$, which means the "collision" would happen 35.1 seconds *before* the second missile is launched. (It is also far below ground level: the equations describe the parabolas extended beyond the actual flights.) So this root is not physical. The first missile is in the air for $2 \cdot 60/10 = 12$ seconds anyway, so it would have landed long before.

For $T = 1$ we get $t = 1.5$ and $s = 0.5 > 0$, so both missiles are in flight. The heights are $60(1.5) - 5(1.5)^2 = 78.75$ and $160(0.5) - 5(0.5)^2 = 78.75$, both positive, and both missiles are at $x = 120$. Hence

$$T = 1 \text{ second}.$$

Checks

Substitute $T = 1$ back in: at $t = 1.5\,\text{s}$ both missiles are at $(120, 78.75)$ metres, which we confirmed with exact fractions. Stepping both motions numerically with launch delay 1 s, the distance between the missiles falls to essentially zero at $t = 1.5\,\text{s}$. As a sanity check, the collision is at $x = 120$, between A and B, and the first missile is still rising, since its vertical velocity there is $60 - 15 = 45\,\text{m}\,\text{s}^{-1}$.

Common slips

- Using the same time variable for both missiles. The second missile has been flying for $t - T$, not t.
- Forgetting that the second missile moves in the negative x-direction.
- Keeping $T = 90$ without checking that $t - T > 0$.

Beyond STEP

There is a slicker way to see this. Both missiles have the same acceleration g, so once both are in flight their *relative* velocity is constant and each sees the other moving in a straight line. A collision happens exactly when that relative velocity points along the line joining them at time T, and writing this down gives the same quadratic. Changing to a freely falling frame like this is the first step towards Einstein's equivalence principle. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats relative motion and non-inertial frames carefully.

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